

Q: What is Python?

A: Python is a high-level, interpreted programming language known for its simple and readable syntax.

Q: What is a list vs a tuple?

A: A list is mutable (can be changed); a tuple is immutable (cannot be changed after creation).

Q: What is a dictionary?

A: A dictionary is a collection of key-value pairs, defined with curly braces: {'key': 'value'}.

Q: What is a lambda function?

A: A lambda is a small anonymous function defined with the lambda keyword, used for short one-liner logic.

Q: What is a decorator?

A: A decorator wraps a function to extend its behaviour without modifying its source code, applied using @syntax.

Q: What is the difference between == and is?

A: == checks if two values are equal; is checks if they refer to the exact same object in memory.

Q: What is a generator?

A: A generator is a function that uses yield to return values one at a time, saving memory for large sequences.

Q: What is list comprehension?

A: A concise one-liner to create a list: [x*2 for x in range(5)] produces [0, 2, 4, 6, 8].

Q: What is the GIL?

A: The Global Interpreter Lock is a mutex in CPython that allows only one thread to run Python code at a time.

Q: What is a context manager?

A: A context manager (used with the with statement) handles setup and teardown of resources like files automatically.